

SCALING SUDOKU AS A CONSTRAINT PROBLEM

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Read each hardness tag from left to right

TAG GROUP	WHAT IT RECORDS
<u>Val</u> · Bnd · Dom	Base propagation: value, bounds-consistent, or domain-consistent
<u>BS</u> · DS	Shaving: bounds shaving or full-domain singleton shaving
<u>Chan</u> · Same · Cmat · <u>Rbm</u>	Additional redundant models and channeling constraints
<u>Search</u>	No tested propagation tag solved the instance without search

Example · ValBS = value propagation + bounds shaving

6×6

PUZZLE

			5		
6					
		2			
	5			6	
	3		2		
				4	

INSTANCE

- Size · 6×6
- Cells · 36
- Hardness · ValChan
- Generation · 0.302 s

9×9

PUZZLE

4						9	7	
			9		4			3
				7	1			
			2			6		
	8		5				2	
6		7						8
	3							7
2		4				3		
	1		8				4	

INSTANCE

- Size · 9×9
- Cells · 81
- Hardness · DomBS
- Generation · 2.212 s

16×16

PUZZLE

2	7		1			C	D	8	G					9	
G				B		4		1		F		3			
				G	7	F		3		5					
	8		2	9	G					1	B		C	A	
				E				9						3	6
1	E	9	7			6	8				B				
6			B	2	5	C				A				1	
						B				4		9		F	D
F		G		4				6	D	A		5		8	
			C			E	A	B	5		8	G			
			E	G					7				4	2	
3		F											2	5	
B					5	A		4		6			C	G	3
9				F								E		6	
		A	4	1	2				E	G					F

INSTANCE

- Size · 16×16
- Cells · 256
- Hardness · Search
- Generation · 33m 12.139s

25x25

PUZZLE

B						1	A	5					0					4		
6					4	P	8		3				F	I	D	7		J	N	C
		3	A	J		H		G	B	E		6	4		M		I	8		A
8	C	5		L	B	F	D			H	I		6	2	K					P
G		P	M		A		2	C			0	J	8					6	F	
3	E	F		5		I			6	8					7					
	M						K	N		H	2	L		D	3			4	9	
0		A		9	D			C				I	5	1	L	F		H	2	
K					G	M	7	P	1				A	6	N	5		F		6
	2	7	1	3	6			0	M		D		I		4	C				5
		N					1		J			H			E	3	6	8		L
			G	2		0	N			5		M	J				3			1
M	5	D				9	L		0	G	F	A		4	C		8	N		P
	J		L					8			I	6		M	9	1		B	H	E
C			4	7	G			5		D	8	N	3			A			9	
E	L		4		D	3				P				6	B		H	2		
				A				G	B	5	I	L			K	P	D		E	0
	3	C			E		4		H		2	9	7		0	N	1		I	6
	F		6							C	1	M		0			3	L	A	N
2		K						7			A	G			C	4		J		3
4	0	E			F				P					8	7			D	B	
	6				P		H			K	J			D		2		G		A
			F			8	1		L				N	P	J		H	5	K	
	A	H	2			4	0			F	7	B				N	9			C
		I	P	G		9					1			0	B	L				M

INSTANCE

- Size · 25x25
- Cells · 625
- Hardness · Search
- Generation · 1h 13m 15.700s

PUZZLE

- Size · 36×36
- Cells · 1,296
- Hardness · Search
- Generation · 1h 27m 55.493s

A tag records the weakest scheme that finishes without search

6 × 6

10 clues · exact local minimum

VAL

		5			3
	1			5	
1			6		4
5					
				2	6

9 × 9

24 clues · exact local minimum

VALBS

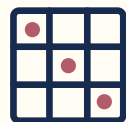
3			9	8		5		
	8		6	7				9
	3							
				5			2	
						4	1	3
6	5			2		8		
4			1					
8			5			9	4	

If no tested scheme finishes by propagation alone, the tag is Search.

Generation and hardness tagging answer different questions

CONSTRUCTION

01



Build grid

one solution

02



Delete clues

smaller batches

03



Keep unique

reject a second

04

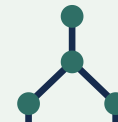


**Check local
minimum**

when certified

ANALYSIS

05



**Tag
hardness**

weakest scheme

06



Walk to Val

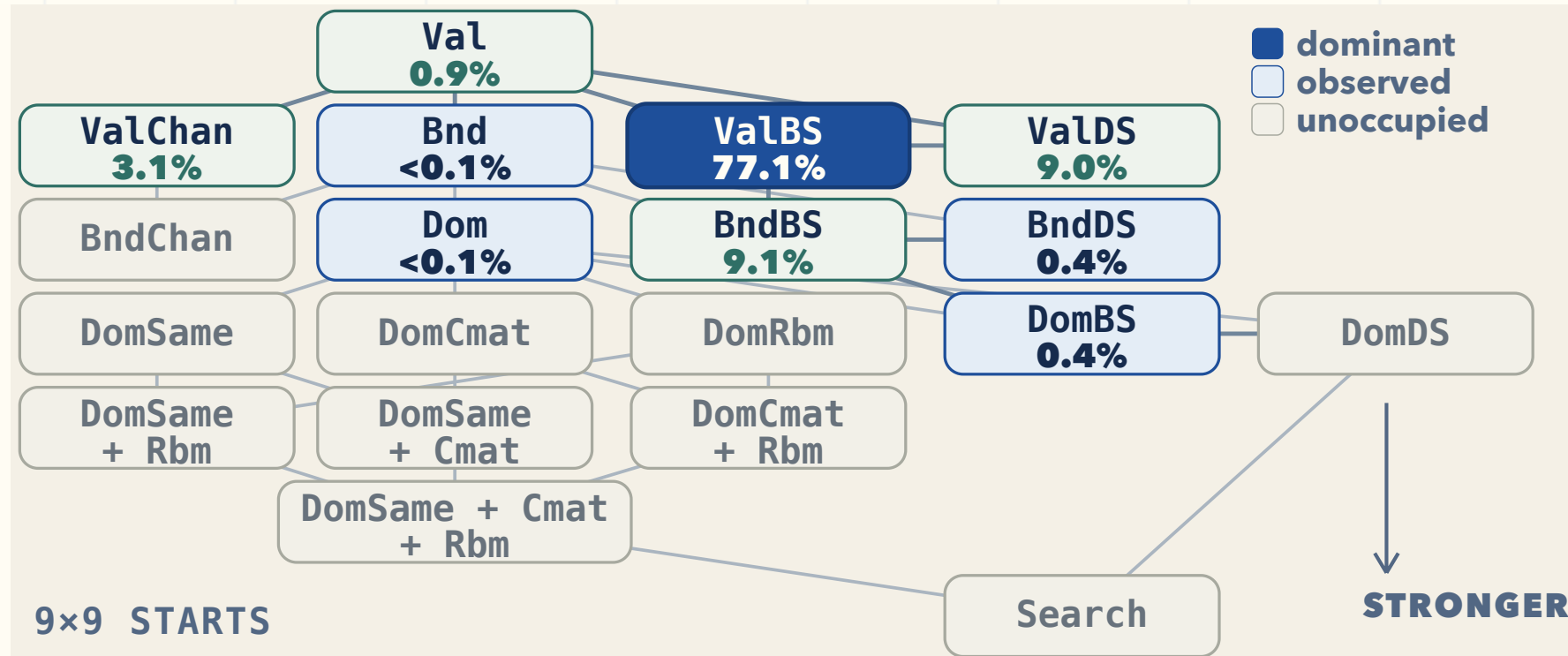
save each drop

434,201 public instances

32,000 bases + 402,201 saved walk variants

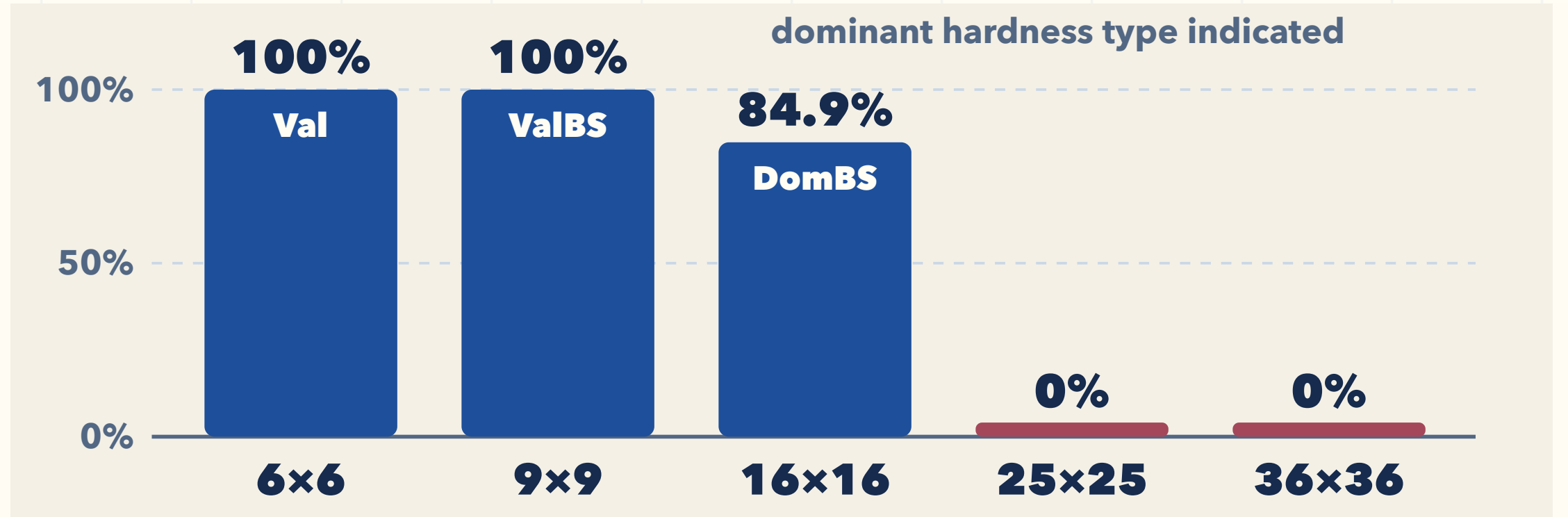
Generate unique puzzles first; measure propagation afterwards.

The 9×9 corpus occupies 8 of 19 propagation schemes



For 9×9 starts, occupied endpoints test an implication; empty nodes provide no test.

Search-free coverage first breaks at 16×16



Coverage falls from 100% at 9×9 to 84.9% at 16×16, then to 0% at 25×25 and 36×36.

Walks: Start from generated 16×16

PUZZLE

			G		6			8		B		A			
7	6				E		4				F	2	8		
				F	B						9	C	G		
	1					3	9	E	5			4			
A			3		5	8			1						6
	D	E		9					C	2					
			9			4	6		8	F		1		G	D
					C		G				E			2	A
8					4		A					5			
	B		1	7	9	C	D		4			F	3		
			2						D		1	9			8
	C					5	F		G						
3		A				F		B			8				
				6	1					7			C		9
F		6		E							5	7		4	2
			B						2	4	C	G			

Dark ink: base hints.

HARDNESS PATH

• ▶ Search · 91 hints · +0

Walks: One added hint reaches DomDS

PUZZLE

			G		6			8		B		A			
7	6				E		4		Ⓐ		F	2	8		
				F	B							9	C	G	
	1					3	9	E	5			4			
A			3		5	8			1						6
	D	E		9					C	2					
			9			4	6		8	F		1		G	D
					C		G				E			2	A
8					4		A						5		
	B		1	7	9	C	D		4			F	3		
			2						D		1	9			8
	C					5	F		G						
3		A				F		B			8				
				6	1					7			C		9
F		6		E							5	7		4	2
			B						2	4	C	G			

HARDNESS PATH

- Search · 91 hints · +0
- ▶ DomDS · 92 hints · +1

Rose circle: new hint on this transition.

Walks: A second hint reaches DomBS

PUZZLE

		G		6		8	B	A	Ⓡ	
7	6			E	4	ⓐ		F	2	8
			F	B				9	C	G
	1				3	9	E	5	4	
A		3		5	8		1			6
	D	E		9			C	2		
			9		4	6	8	F	1	G
				C		G		E		2
8				4	A				5	
	B		1	7	9	C	D		F	3
			2				D	1	9	
	C				5	F	G			8
3		A			F	B		8		
				6	1			7		C
F		6		E				5	7	4
			B				2	4	C	G

HARDNESS PATH

- Search · 91 hints · +0
- DomDS · 92 hints · +1
- ► DomBS · 93 hints · +2

Blue square: earlier addition · rose circle: new.

Walks: Ten more hints reach BndDS

PUZZLE

			G		6			8		B		A		F
7	6				E		4		A		F	2	8	
		⑧		F	B			②			9	C	G	
	1			⑧		3	9	E	5			4		
A			3		5	8			1					6
⑥	D	E		9			①		C	2				
			9			4	6		8	F		1	G	D
		①			C		G				E		2	A
8					4		A	⑨			⑤	5		
	B		1	7	9	C	D		4			F	3	
			2						D		1	9		8
	C					5	F		G					
3		A				F		B			8			
				6	1					7			C	9
F		6		E				①		5	7		4	2
			B				⑧		2	4	C	G		

HARDNESS PATH

- Search · 91 hints · +0
- DomDS · 92 hints · +1
- DomBS · 93 hints · +2
- ► BndDS · 103 hints · +12

Previous additions stay boxed; this transition is circled.

Walks: One more hint reaches BndBS

PUZZLE

			G		6			8		B		A		F
7	6				E		4		A		F	2	8	
		8		F	B			2			9	C	G	
	1			8		3	9	E	5			4		
A			3		5	8			1					6
G	D	E		9			1		C	2				
			9			4	6		8	F		1	G	D
		1			C		G				E		2	A
8					4		A	9			B		5	
	B		1	7	9	C	D		4			F	3	
			2						D		1	9		8
	C					5	F		G					
3		A				F		B			8			
				6	1				E	7			C	9
F		6		E				1			5	7		4
			B				8		2	4	C	G		2

HARDNESS PATH

- Search · 91 hints · +0
- DomDS · 92 hints · +1
- DomBS · 93 hints · +2
- BndDS · 103 hints · +12
- ► BndBS · 104 hints · +13

The shape change identifies the newest hint without color.

Walks: Eleven more hints reach ValBS

PUZZLE

			G		6			8		B		A		F
7	6		5		E		4		A		F	2	8	
		8	E	F	B			2			9	C	G	
	1	2		8		3	9	E	5			4	7	
A			3		5	8		G	1					6
G	D	E		9			1		C	2				
			9			4	6		8	F		1		G D
		1			C		G	4			E		9	2 A
8					4		A	9			B		5	
6	B		1	7	9	C	D		4	8		F	3	
			2						D		1	9		8
	C					5	F		G					
3		A		4		F		B			8			
				6	1				E	7			C	9
F		6		E				1			5	7		4 2
		7	B				8		2	4	C	G		

HARDNESS PATH

- Search · 91 hints · +0
- DomDS · 92 hints · +1
- DomBS · 93 hints · +2
- BndDS · 103 hints · +12
- BndBS · 104 hints · +13
- ► ValBS · 115 hints · +24

The path is cumulative; only the newest delta is circled.

Walks: Forty added hints finally reach Val

PUZZLE

			G		6		8		B		A		F	
7	6		5		E	1	4		A		F	2	8	
4		8	E	F	B			2			9	C	G	
	1	2		8		3	9	E	5			4	7	
A			3	1	5	8		G	1					6
G	D	E		9			1		C	2				
			9			4	6	5	8	F		1		G
	F	1	6		C		G	4			E		9	2
8					4		A	9		B	6	5		
6	B		1	7	9	C	D		4	8	2	F	3	
			2					D		1	9			8
	C					5	F	G			1			
3		A	1	4		F		B			8	E		5
			4	6	1			E	7			C		9
F	6			E				1		A	5	7	4	2
1		7	B		1	8			2	4	C	G		

HARDNESS PATH

- Search · 91 hints · +0
- DomDS · 92 hints · +1
- DomBS · 93 hints · +2
- BndDS · 103 hints · +12
- BndBS · 104 hints · +13
- ValBS · 115 hints · +24
- ► Val · 131 hints · +40

40 cumulative additions: 24 boxed earlier · 16 circled now.

The 434,201-instance corpus

GRID	BASES	WALKS	SAVED VARIANTS	EXACT LOCAL MINIMALITY
6×6	10,000	50,000	49,515	100.0%
9×9	10,000	50,000	92,456	100.0%
16×16	10,000	50,000	167,026	100.0%
25×25	1,000	10,000	46,079	100.0%
36×36	1,000	10,000	47,125	0.2%

Corpus total · 32,000 bases + 402,201 saved walk variants
= 434,201 instances. Exact local minimality is certified for
only 0.2% of the 36×36 bases.

9×9 checks the setup; 16×16 separates propagation schemes

○ **9×9 · SANITY CHECK**

100%

10,000 bases

Mostly one tag

□ **16×16 · DISCRIMINATOR**

84.9%

10,000 bases

Several tags + 15.1% Search

[Consequence · Start with tagged 16×16. Add 25×25 and 36×36 when the question is size scaling or missing reasoning.

Scale changes what Sudoku benchmarks measure

9×9 is simple.
16×16 and
up more
interesting.



github.com/zayenz/scaled-sudoku-instances

434,201 instances to explore.